

INTERNSHIP REPORT

on

Building a Beginner-Friendly Chatbot

Submitted as part of the Machine Learning Internship Program
at Codec Technologies Pvt. Ltd.

Submitted By

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To school of computer science engineering



IILM UNIVERSITY GRATER NOIDA

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AICTE & ICAC Approved Internship Program — National Internship Portal

Internship Details

Student Name	Mayank Kumar
Designation	Machine Learning Intern (Artificial Intelligence Intern)
Organization	Codec Technologies Pvt. Ltd.
Duration	29 June 2026 to 29 July 2026 (1 Month)
Mode	Pan India, Hybrid
Program	AICTE & ICAC Approved Internship, National Internship Portal
Project Title	Design and Development of a Simple Rule-Based Chatbot

1. Acknowledgement

I would like to express my sincere gratitude to Codec Technologies Pvt. Ltd. for giving me the opportunity to undergo this internship and work on a live, hands-on project. I am thankful to Dr. Anurag Shrivastava and the project mentors for their guidance and support throughout the program. This internship, conducted under the AICTE & ICAC approved program on the National Internship Portal, helped me gain valuable practical exposure to Machine Learning and Artificial Intelligence concepts that go beyond classroom learning.

2. Company Profile

Codec Technologies Pvt. Ltd. is a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. The company delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth. It is a recognized Google for Education partner and is ISO 9001:2015 certified, and it runs its internship programs through the National Internship Portal under the Ministry of Education.

3. Introduction

This report summarizes the work carried out during a one-month Machine Learning internship at Codec Technologies Pvt. Ltd. The internship, conducted from 29 June 2026 to 29 July 2026 in hybrid mode, provided hands-on exposure to how a real AI/ML project is planned, built and demonstrated. As part of the internship, a beginner-level chatbot was designed and implemented to understand the fundamentals of conversational AI and natural language processing (NLP). The sections below walk through the problem statement, objectives, tools used, methodology, results and key learnings from the project.

4. Problem Statement

Most beginners find it difficult to understand how conversational AI systems work in practice. There was a need for a simple system that could take a user's text message, understand its basic intent, and reply automatically — without relying on complex, resource-heavy AI models that are hard for a newcomer to build, debug or explain. The project addresses this by building a lightweight chatbot that responds instantly to common greetings and questions, is easy to set up on a personal computer, and is simple enough to explain step by step in a review or interview.

5. Weekly Work Summary

The one-month internship was structured week by week, moving from learning the basics to building and finishing the chatbot project:

Week 1	Onboarding, orientation and introduction to Machine Learning and NLP basics. Explored the internship platform and finalized the chatbot project idea.
Week 2	Learned core Python concepts needed for the project and studied how simple rule-based chatbots work.
Week 3	Implemented the chatbot logic — text cleaning, keyword matching and response selection — and tested it with sample conversations.
Week 4	Refined the keyword list and responses, fixed issues found during testing, and prepared the final project report and presentation.

6. Objectives

- Build a simple, rule-based chatbot using Python.
- Enable the chatbot to understand common greetings, questions and keywords.
- Generate relevant, automatic replies to the user in real time.
- Learn the basic building blocks of NLP and conversational AI.
- Deliver a working, demo-ready beginner project by the end of the internship.

7. Tools and Technologies Used

- Python — core programming language used to write the chatbot logic.
- Keyword / rule-based matching — simple if-else and keyword-matching logic to understand user input.
- Basic string handling (NLP basics) — lowercasing, cleaning and matching text.
- VS Code / Jupyter Notebook and Google Colab — environments used to write, run and test the chatbot.
- Command line interface — the chatbot runs and converses with the user directly in the terminal/console.

8. Methodology

The chatbot was developed in five simple steps that mirror a typical beginner NLP workflow:

- Collect sample inputs — listed common things users might type, such as greetings, questions, thank-yous and goodbyes.
- Clean the text — converted user input to lowercase and removed extra spaces/punctuation so matching is reliable.
- Match keywords — compared the cleaned input against predefined keyword lists using simple if-else conditions.
- Pick a response — for each matched keyword group, the program selects a suitable reply from a small response set.
- Chat in a loop — the whole process runs inside a loop so the bot keeps chatting until the user types “bye”.

For example, if the input contains “hello” or “hi”, the bot replies with a greeting; if it contains “bye”, the conversation loop ends.

9. System Flow

The overall flow of the chatbot can be summarized as a straightforward, five-stage pipeline that runs every time the user sends a message:

- Stage 1: User types a message into the console.
- Stage 2: The text is cleaned (lowercased and stripped of extra characters).
- Stage 3: The cleaned text is checked against the keyword lists.
- Stage 4: The best matching reply is selected from the response set.
- Stage 5: The bot displays the reply and waits for the next message.

This simple, linear flow makes the chatbot easy to trace, debug and explain to others, which was an important consideration given the beginner scope of the project.

10. Results and Sample Output

The chatbot successfully responds to common conversational inputs in real time. A sample run of the console-based chatbot is shown below:

User	Hi there!
Bot	Hello! How can I help you today?
User	What is your name?
Bot	I'm a simple chatbot built by Mayank.
User	Thank you
Bot	You're welcome! Happy to help.
User	Bye
Bot	Goodbye! Have a great day.

Key features observed: the chatbot understands greetings and basic questions, gives instant automated replies, runs continuously until the user says “bye”, is lightweight with no heavy libraries required, and can easily be extended with more keywords in the future.

11. Challenges Faced

- Handling the different ways users phrase the same question.
- Avoiding situations where the bot gives no reply for unmatched input.
- Keeping the keyword lists organized and readable as they grew larger.

12. Learning Outcomes

- Understood the basics of how chatbots and NLP systems work.
- Practiced writing clean, structured Python logic.
- Gained experience debugging and testing conversational flows.
- Learned how to present a technical project in a simple, easy-to-explain manner.

13. Future Scope

While the current chatbot is intentionally simple, it can be extended in several ways as the next step in learning:

- Add more keywords and response categories to widen the range of topics the bot can handle.
- Use basic NLP libraries such as NLTK to handle spelling variations and sentence structure better.
- Introduce a similarity-based matching approach (e.g. TF-IDF) instead of exact keyword matching.
- Build a simple graphical or web-based interface instead of the command line.
- Connect the chatbot to a small knowledge base so it can answer more specific, domain-related questions.

14. Conclusion

The one-month internship at Codec Technologies Pvt. Ltd. provided valuable, hands-on exposure to the practical side of building AI/ML applications. Through the design and development of a simple, rule-based chatbot, the fundamentals of natural language processing, Python programming and conversational system design were learned and applied. The project met all its stated objectives and resulted in a working, demo-ready chatbot that can be explained step by step, forming a strong foundation for exploring more advanced machine learning and NLP techniques in the future.

Certificate of Internship and Offer Letter issued by Codec Technologies Pvt. Ltd. are attached as supporting documents.